Highlights

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Characterization Study

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- Decoder contribution is spatially heterogeneous and highly concentrated.
- Efficient decoders achieve net-neutral voxel correctness at matched parameters.
- Uncertainty signals predict where predictions flip, but not improvement direction.
- Decoder redundancy is structured, enabling future adaptive inference strategies.

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Characterization Study

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Abstract

Recent advancements in U-Net based 3D medical image segmentation have introduced increasingly computation-heavy decoders, particularly with the integration of attention and Transformer-based modules. In response, efficient models such as EffiDec3D simplify or remove these decoders based on empirical end-point tradeoffs and the assumption that static decoder capacity is globally redundant. However, these efficiency-focused approaches fail to answer a fundamental scientific question: where, how much, and under what conditions is this heavy decoder computation actually useful? To fill this gap, we present a systematic characterization study of decoder computation. By evaluating voxel-level disagreements between matched full and efficient decoders across representative architectures and datasets, we characterize decoder utility in terms of heterogeneity, concentration, and predictability. Our results demonstrate that decoder computation is spatially heterogeneous; the full decoder provides a statistically net-neutral benefit, with improvements counterbalanced by degradations. However, this net-neutral effect is highly concentrated in a thin boundary band, revealing structured redundancy rather than uniform irrelevance. Finally, we show that while cheap inference-time uncertainty signals reliably predict the location of prediction instability, they fail to predict the direction of improvement, rendering simple selective-allocation strategies ineffective. Ultimately, our findings reveal

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that modern decoder computation exhibits structured redundancy, providing the scientific constraints necessary for designing future adaptive and dynamic decoding architectures.

Keywords: 3D Medical Image Segmentation, Efficient Deep Learning, Uncertainty, Decoder Redundancy, Model Characterization

1. Introduction

Precise 3D medical image segmentation is critical for clinical applications such as disease screening, surgical planning, and image-guided intervention. To meet these demands, deep learning architectures have historically relied on the encoder-decoder paradigm established by U-Net [1, 2]. Over the past several years, the field has witnessed a continuous evolution toward increasingly expressive encoder representations. Concurrently, modern architectures—including UNETR [3], Swin UNETR [4], 3D UX-Net [5], and MedNeXt [6]—have largely retained dense, multi-resolution decoding stages. More recent designs further enhance decoder capability through advanced feature fusion, refinement, and upsampling modules. This trajectory reflects an implicit assumption that heavy decoder computation is universally beneficial for accurate segmentation.

Despite these continuous advances, existing literature evaluates decoder design almost exclusively through aggregate, end-point metrics such as the Dice similarity coefficient or Hausdorff distance. Such global metrics offer a macroscopic view of model performance and efficiency trade-offs, but they critically obscure the micro-level behavior of the network. Specifically, they fail to answer a fundamental scientific question: *Where, how much, and under what conditions is additional decoder computation actually useful?*

To address this gap, we present a systematic characterization study of decoder computation in 3D medical image segmentation. Rather than introducing a new architecture, we employ matched pairs of full and efficient decoders (specifically building upon the EffiDec3D framework [7]) as controlled experimental instruments. By isolating the decoder capacity while holding the encoder, data, and training protocols constant, we directly observe the voxel-level disagreements—the "flips"—caused by adding decoder capacity.

Our characterization framework yields the following core contributions that challenge the prevailing assumption of uniform decoder utility:

- **Heterogeneous Contribution**: We demonstrate that the marginal benefit of a full decoder over an efficient baseline is almost entirely localized to thin boundary bands and small, difficult anatomical structures.

- **Concentrated Benefit**: Our oracle analysis reveals that the vast majority of the full decoder's positive corrections are clustered in a small fraction of the image volume ($< 5\%$), providing direct evidence of structured decoder redundancy.

- **Predictability of Instability**: We show that while lightweight inference-time uncertainty signals (e.g., entropy) can reliably predict *where* predictions will be unstable, they fail to predict the *direction* of improvement.

- **Net-Neutral Voxel Correctness**: At the voxel level, the matched full decoder's net benefit is statistically indistinguishable from zero, representing a net-neutral effect where positive corrections are counterbalanced by negative degradations.

Collectively, these findings demonstrate that modern decoder computation exhibits a structured pattern of heterogeneous contribution, concentrated benefit, and partially predictable instability, providing the scientific foundation for future adaptive and dynamic decoding strategies. The rest of this paper is organized as follows: Section 2 reviews related work, Section 3 details our characterization framework, Section 4 presents the experimental setup and results, Section 5 discusses the implications, and Section 6 concludes the paper.

2. Related Work

2.1. Decoder Design in 3D Medical Image Segmentation

U-Net [1, 2] and V-Net [8] established the dominant encoder-decoder paradigm for 3D medical image segmentation, characterizing dense, multi-stage decoding and skip connections, which culminated in self-configuring frameworks like nnU-Net [9]. Subsequent architectural advancements have consistently evolved toward increasingly expressive encoder representations and hybrid transformer backbones, such as TransUNet [10]. Modern 3D networks, including UNETR [3], Swin UNETR [4], 3D UX-Net [5], and MedNeXt [6], have substantially strengthened feature extraction while largely retaining dense multi-resolution decoder designs. More recent architectures further enhance decoder capability through improved upsampling, feature fusion, and feature refinement mechanisms.

Despite these continuous advances in decoder design, prior work evaluates decoders almost exclusively through final segmentation accuracy and aggregate computational cost. Little work investigates *where* additional decoder computation contributes spatially, or whether its contribution is uniform across the target anatomy.

2.2. Efficient Decoder Design

In response to the computational bottleneck of 3D segmentation, recent works have begun to explicitly target decoder efficiency. The EffiDec3D framework [7] introduced a systematic approach to decoder simplification through channel reduction and the removal of high-resolution decoder stages, establishing a strong architecture-level efficiency-accuracy tradeoff. Similarly, EfficientMedNeXt [11] adopted comparable decoder simplifications before introducing specialized convolution blocks.

Other lightweight segmentation architectures improve efficiency through general architectural redesign, lightweight attention mechanisms, or hybrid encoders (e.g., SegFormer [12], UNETR++ [13], Slim UNETR [14]) rather than explicitly reducing or characterizing decoder computation. Crucially, existing efficient decoder studies infer redundancy implicitly from end-point accuracy-efficiency trade-offs. They do not directly characterize where decoder computation contributes at the voxel level.

2.3. Characterization Studies

Existing characterization studies in medical image segmentation primarily investigate prediction uncertainty, calibration, prediction reliability, failure detection, and segmentation quality prediction using techniques such as Monte Carlo Dropout [15] and Deep Ensembles [16] (e.g., Jungo & Reyes [17], CMIG 2025 studies [18, 19], and comprehensive reviews [20]). These studies often evaluate how well uncertainty metrics correlate with overall error rates or how they can be used to flag uncertain cases for clinician review.

Our work fundamentally diverges from this literature. Rather than characterizing general prediction uncertainty, we characterize the marginal contribution of decoder computation itself. Prior work has reduced decoder computation or studied prediction uncertainty, but has not systematically characterized the utility of decoder capacity through voxel-level heterogeneity, benefit concentration, and inference-time predictability.

3. Methods: A Characterization Framework

To understand the spatial and functional utility of decoder computation, we design a controlled observation study. We treat a matched full decoder and an efficient decoder as experimental instruments, isolating the effect of decoder capacity on voxel-level predictions.

3.1. Experimental Setup

Let v denote a voxel within a 3D medical image. We evaluate two decoder configurations sharing identical encoder features, training protocols, and data splits:

- **Efficient Decoder (E_1)**: Parameter-matched efficient decoder based on the EffiDec3D configuration, utilizing concatenation for skip aggregation to strictly align with published parameter counts.
- **Full Decoder (E_0)**: The standard dense, high-resolution decoder of the respective backbone architecture.

For a given voxel v, let $y(v)$ be the ground truth label, $\hat{y}_e(v)$ be the prediction from the efficient decoder, and $\hat{y}_f(v)$ be the prediction from the full decoder.

3.2. Voxel-level Flip Analysis

Rather than relying on aggregate metrics like Dice, which sum all errors indiscriminately, we track how the full decoder alters the prediction of the efficient baseline. We define three categories of prediction "flips" when transitioning from the efficient to the full decoder:

- **Positive Flip ($P(v)$)**: The full decoder improves a previously incorrect prediction. $P(v) = 1[\hat{y}_e(v) \neq y(v) \wedge \hat{y}_f(v) = y(v)]$.
- **Negative Flip ($N(v)$)**: The full decoder degrades a previously correct prediction. $N(v) = 1[\hat{y}_e(v) = y(v) \wedge \hat{y}_f(v) \neq y(v)]$.

- **Net Benefit ($U(v)$)**: The voxel-wise difference between improvements and degradations. $U(v) = P(v) - N(v)$.

The global rates are defined as $R_{pos} = \text{mean}_v P(v)$, $R_{neg} = \text{mean}_v N(v)$, and $R_{net} = R_{pos} - R_{neg}$.

3.3. Core Metrics

We operationalize our investigation into structured redundancy through the following specific metrics:

1. Global Net Benefit: This metric evaluates the macro-level necessity of the full decoder. The global net flip rate R_{net} is computed as the expected voxel-wise net benefit over the volume V:

$$R_{net} = \frac{1}{|V|} \sum_{v \in V} U(v) \tag{1}$$

Its physical meaning answers whether the full decoder provides a unidirectional advantage ($R_{net} > 0$) or if its positive corrections and negative degradations cancel each other out ($R_{net} \approx 0$).

2. Boundary and Anatomy Resolution: To evaluate spatial heterogeneity, we analyze the net benefit $U(v)$ as a function of the Euclidean distance to the nearest organ boundary, as well as its distribution across different anatomical structures. This reveals *where* the decoder is actually intervening.

3. Oracle Concentration: This metric measures the spatial sparsity of the decoder's utility. We divide the image into 3D blocks b_r and compute the net benefit $B(r) = \sum_{v \in b_r} U(v)$ for each block. We rank the blocks by $B(r)$ and calculate the fraction of the total positive net mass recovered by the top $k\%$ of blocks:

$$Coverage(k) = \frac{\sum_{r=1}^{k\% \times |B|} \max(B(r), 0)}{\sum_{r=1}^{|B|} \max(B(r), 0)} \tag{2}$$

We summarize this curve using the K_{80} metric, which is the execution budget required to recover 80% of the full decoder's positive benefit.

4. Flip Predictability: This evaluates whether we can forecast the decoder's behavior before executing it. We use lightweight inference-time signals from the efficient decoder, such as predictive entropy:

$$H(v) = -\sum_{c=1}^{C} p_c(v) \log p_c(v) \tag{3}$$

where $p_c(v)$ is the softmax probability for class c. We measure predictability using the Area Under the Receiver Operating Characteristic (AUROC) to predict the *location* of flips (instability) and the *direction* of flips (positive vs. negative).

5. Selection Gap: This measures the practical, deployable opportunity for selective computation. We compare the net benefit recovered by an uncertainty-based block selector (e.g., selecting blocks with the highest mean entropy) against a random selection policy at a fixed computational budget (e.g., 20%). A significant positive gap indicates a viable routing strategy.

4. Experiments and Results

4.1. Experimental Setup

We evaluate matched decoder pairs across representative datasets to answer three core questions: (1) Is the full decoder's contribution spatially uniform or heterogeneous? (2) Is the benefit broadly distributed or concentrated in specific regions? (3) Can we predict where the full decoder will improve predictions using cheap inference-time signals?

To ensure cross-architecture generality, our characterization study evaluates three distinct architectural paradigms: a large-kernel CNN (3D UX-Net [5]), a vision transformer backbone (SwinUNETR [4]), and a modern ConvNeXt-style medical network (MedNeXt_M [6]). Experiments are conducted on the BTCV multi-organ CT dataset (13 organs) [21] and the FeTA 2021 fetal brain MRI dataset (7 structures) [22].

Hardware and Environment: All models were trained on an AutoDL RTX 5090 (32 GB VRAM, Blackwell architecture) utilizing CUDA 12.8 and PyTorch 2.6. We employed BF16 mixed precision to accelerate training.

Training Protocol: Both the full decoder models (E_0) and the efficient baseline models (E_1) were trained for a total of 45,000 optimizer steps with identical data augmentations, learning rates (0.001), and a crop size of $96 \times 96 \times 96$. Model evaluation was performed every 250 steps.

Reproduction and Baseline Comparison: Table 1 presents the aggregate computational complexity and segmentation performance across all three backbones. Our efficient decoder implementation (E_1) strictly matches the published parameter reduction strategy of EffiDec3D [7]. Across all three backbones, E_1 achieves massive FLOP reductions (up to 91.4% in 3D UX-Net) while retaining most of the baseline segmentation accuracy. For instance, MedNeXt_M E_1 maintains an 81.47% Mean Dice (vs. 83.74% for

Table 1: Main quantitative comparison of Full (E_0) vs. Efficient (E_1) decoders on the BTCV 13-organ dataset across three backbones.

Architecture	Variant	Params (M) ↓	FLOPs (GMac) ↓	Latency (ms) ↓	Mean Dice (%) ↑	Mean HD95 (mm) ↓
3D UX-Net	Full (E_0)	53.01	578.74	40.6	79.18	9.04
	Effi (E_1)	3.16	49.49	8.0	77.73	11.82
SwinUNETR	Full (E_0)	62.19	308.24	37.0	80.48	8.88
	Effi (E_1)	11.21	55.84	16.7	79.49	8.90
MedNeXt_M	Full (E_0)	39.32	230.87	40.9	83.74	5.77
	Effi (E_1)	12.95	106.91	18.7	81.47	6.60

Table 2: 2×2 Factorial decomposition of decoder channel reduction (C_{dec}) and resolution scaling (RF) on 3D UX-Net (BTCV dataset).

Configuration	C_{dec}	RF	Params (M)	FLOPs (GMac)	Latency (ms)	Mean Dice (%)
Full (C_{384}, RF_1)	384	1	46.72	657.76	43.3	80.44
Chan-Only (C_{48}, RF_1)	48	1	2.24	388.15	35.2	79.80
Res-Only (C_{384}, RF_2)	384	2	46.32	319.10	16.1	77.97
Effi (C_{48}, RF_2)	48	2	1.84	49.49	8.0	78.35

E_0) while reducing inference latency from 40.9 ms to 18.7 ms. Similarly, SwinUNETR E_1 achieves 79.49% Mean Dice (vs. 80.48% for E_0) while cutting FLOPs from 308.24 GMac to 55.84 GMac. This tight control over encoder weights, data split, and training regimen isolates the decoder capacity for our voxel-level characterization.

4.2. Factorial Factor Analysis

To dissect whether decoder redundancy stems primarily from channel over-parameterization (C_{dec}) or spatial resolution scaling (RF), we conduct a 2×2 factorial ablation on 3D UX-Net (Table 2). Reducing decoder channels from 384 to 48 (C_{48}, RF_1) compresses parameters from 46.72M to 2.24M while maintaining 79.80% Mean Dice. Conversely, removing the high-resolution stage (C_{384}, RF_2) reduces latency to 16.1 ms. Combining both (C_{48}, RF_2) yields the ultra-lightweight E_1 configuration (1.84M params, 49.49 GMac), proving that channel reduction and resolution scaling offer orthogonal efficiency dimensions.

4.3. Heterogeneity: The Net-Neutral Boundary Band

We evaluate the overall voxel-level net benefit of the full decoder across all three backbones. Surprisingly, with parameter-matched efficient configurations, the full decoder's net benefit is statistically net-neutral across all architectures. For 3D UX-Net on BTCV, the subject-mean net flip rate is

$+0.046\%$ (95% CI: $[-0.0002\%, +0.0011\%]$). For SwinUNETR, the net flip rate is -0.0008% (95% CI: $[-0.044\%, +0.039\%]$), and for MedNeXt_M, it is $+0.085\%$ (95% CI: $[+0.024\%, +0.152\%]$). As shown in Figure 2, across 3D UX-Net, SwinUNETR, and MedNeXt_M, positive corrections are counterbalanced by negative degradations, establishing a universal net-neutral voxel correctness.

Furthermore, this net-neutral behavior is highly heterogeneous. The boundary error rate is $3.93\times$ higher than the interior error rate in 3D UX-Net and $4.67\times$ higher in MedNeXt_M. In Figure 2, net flip activity across all networks is tightly concentrated in a thin boundary band (0-1 voxels from organ edges). Figure 3 details the organ-level flip rates across all three networks. Small and morphologically complex structures experience high flip volatility across all architectures: Left/Right Adrenal glands exhibit positive flip rates of 16.2%/16.9% in 3D UX-Net, 14.4%/14.8% in SwinUNETR, and 18.3%/10.4% in MedNeXt_M. In contrast, large organs like the liver remain highly stable ($< 3\%$ flip activity across all models).

4.4. Concentration: Structured Redundancy

We investigate the spatial concentration of decoder benefit using an oracle block-selector across all three networks. Figure 4 presents the oracle coverage curves for 3D UX-Net, SwinUNETR, and MedNeXt_M. The curves are extremely steep across all architectures: an oracle selecting only the top 5% of blocks recovers 80% of the full decoder's positive net mass ($K_{80} = 5\%$) in 3D UX-Net, SwinUNETR, and MedNeXt_M alike. By a 10% budget, oracle coverage reaches 99.98%, and 100% at 20%. This stark concentration across all three networks proves that the vast majority of the image volume derives no meaningful benefit from the heavy decoder layers.

4.5. Predictability: Instability vs. Improvement

We evaluate whether lightweight inference-time signals (predictive entropy and confidence) can identify these beneficial regions prior to executing the full decoder.

Across all three backbones (Figure 5), our results reveal a stark dichotomy between instability and directional improvement: - **3D UX-Net**: Flip location (instability) AUROC is 0.903 (95% CI: $[0.876, 0.926]$), but flip direction AUROC is 0.592 (95% CI: $[0.560, 0.626]$). - **SwinUNETR**: Flip location AUROC is 0.912 (95% CI: $[0.884, 0.934]$), but flip direction AUROC is 0.561 (95% CI: $[0.512, 0.608]$). - **MedNeXt_M**: Flip location AUROC is 0.931 (95% CI: $[0.917, 0.944]$), but flip direction AUROC is 0.545 (95% CI: $[0.526, 0.567]$).

Consequently, the deployable selection gap is zero across all backbones. At a 20% budget, entropy-based block selection yields net benefits statistically indistinguishable from a random policy across 3D UX-Net, SwinUNETR, and MedNeXt_M, proving that simple uncertainty thresholding cannot reliably recover the theoretical efficiency opportunity.

5. Discussion

The results of our characterization study compel a re-evaluation of how decoder computation is allocated and assessed in 3D medical image segmentation. The central finding—that the marginal benefit of a full decoder is net-neutral at the voxel level and tightly clustered around organ boundaries—challenges the necessity of dense, uniform decoding for the entirety of an input volume.

Cross-Architecture Generality: Crucially, our empirical observations hold remarkably consistent across three fundamentally different architectural paradigms: a large-kernel CNN (3D UX-Net), a vision transformer (SwinUNETR), and a modern ConvNeXt-style medical model (MedNeXt_M). Regardless of whether the encoder relies on local convolutions, global self-attention, or large-depthwise kernels, decoder capacity exhibits the same structured redundancy: net-neutral voxel correctness, $K_{80} = 5\%$ oracle concentration, and high flip-location predictability paired with chance-level direction predictability.

Observation vs. Action: Our study clearly delineates the boundary between observation and action. The high AUROC for flip location (> 0.90 across all networks) confirms that regions where predictions change are deterministic and easily identified by cheap uncertainty signals. High entropy strongly correlates with boundary regions, exactly where the decoder acts. However, the inability to predict the *direction* of these flips (AUROC ≈ 0.55) represents a significant negative result for simple region-routing strategies. Because the baseline efficient decoder is already highly capable, deploying the full decoder on uncertain regions is just as likely to disturb a correct prediction as it is to fix an incorrect one.

This implies that we cannot realize the theoretical computational savings of structured redundancy merely by selectively gating the execution of existing static decoders using uncertainty thresholds. The predictability of instability (but not improvement) necessitates a shift from attempting to "turn off" decoder layers to developing genuinely adaptive architectures that dynamically allocate capacity based on feature complexity rather than simple confidence scores.

6. Conclusion

In this work, we presented a systematic characterization study of decoder computation in modern 3D medical image segmentation architectures. By analyzing the voxel-level differences between matched full and efficient decoders, we demonstrated that modern decoder computation exhibits structured redundancy: its contribution is highly heterogeneous, its benefits are spatially concentrated at object boundaries, and the instability of its predictions is easily detectable.

However, we also showed that the directional utility of this computation is currently unpredictable using standard inference-time signals. The full decoder is net-neutral in voxel correctness, revealing that simple uncertainty-based routing is insufficient to harness this redundancy. This work establishes the inherent limitations of static decoding and naive thresholding. Future work will leverage these constraints to design genuinely dynamic architectures that adaptively allocate decoder computation based on local feature complexity, moving beyond simple confidence scores to harness this structured redundancy.

CRediT authorship contribution statement

Author One: Conceptualization, Methodology, Writing – original draft. **Author Two**: Data curation, Software. **Author Three**: Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study (BTCV/Synapse, FeTA 2021) are publicly available.

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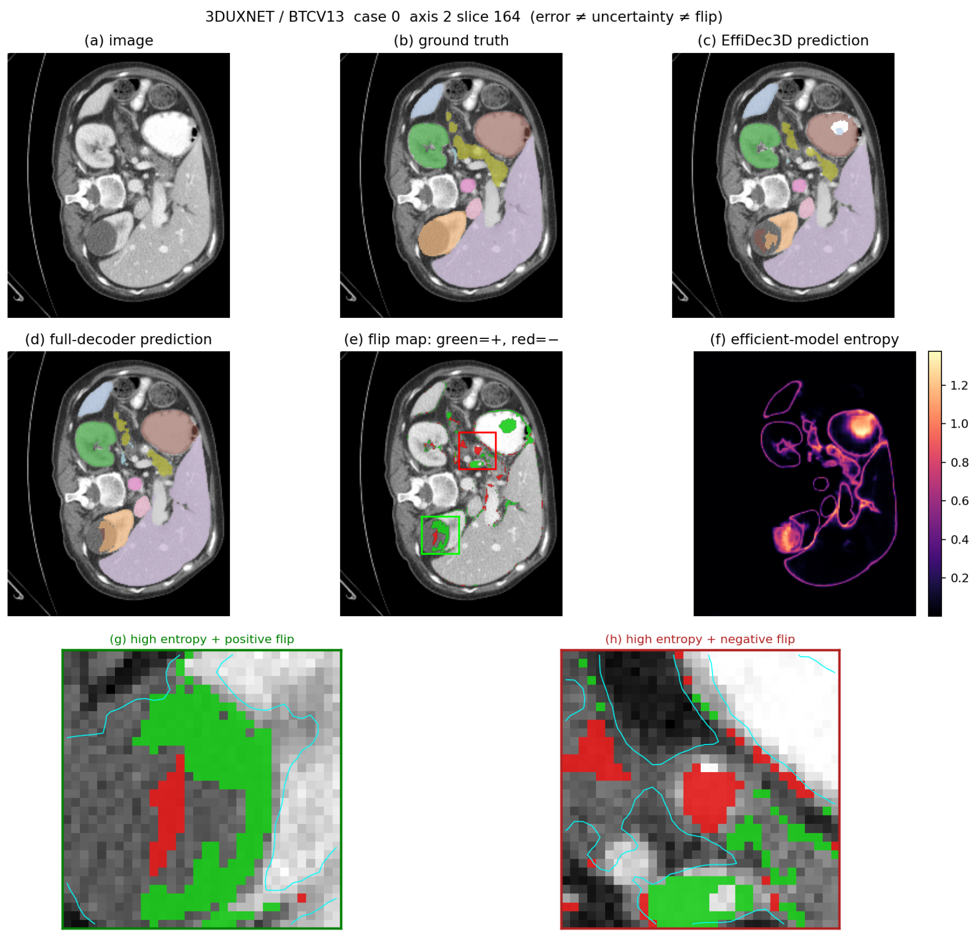
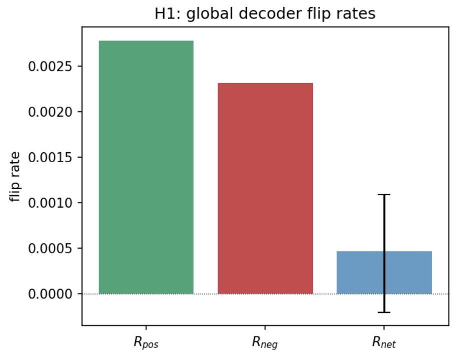
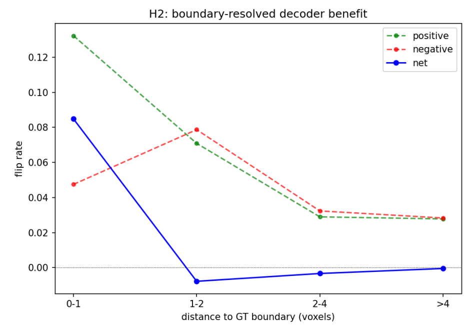


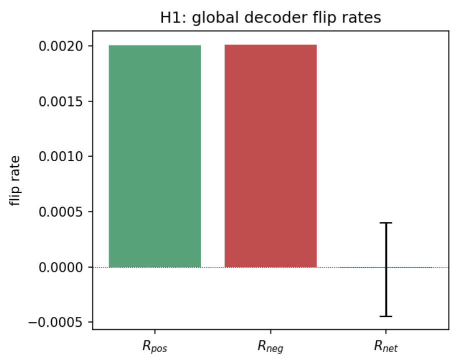
Figure 1: Motivation: Decoder computation yields spatially localized flips. While the full decoder corrects some efficient decoder errors (positive flips), it simultaneously degrades other previously correct predictions (negative flips) at high-entropy boundaries. This illustrates that decoder computation is not uniformly beneficial.



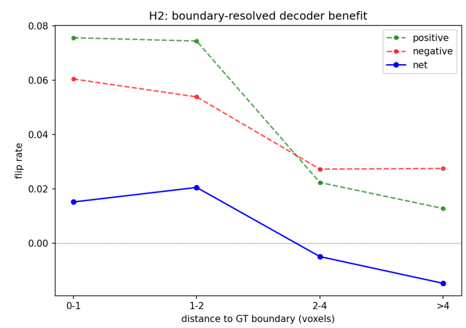
(a) 3D UX-Net Global Flips



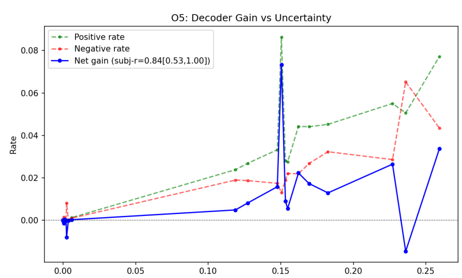
(b) 3D UX-Net Boundary Flips



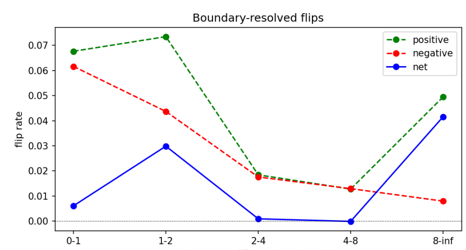
(c) SwinUNETR Global Flips



(d) SwinUNETR Boundary Flips

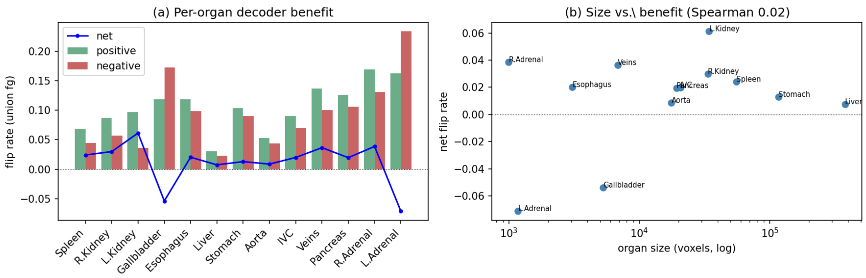


(e) MedNeXt_M Global Flips

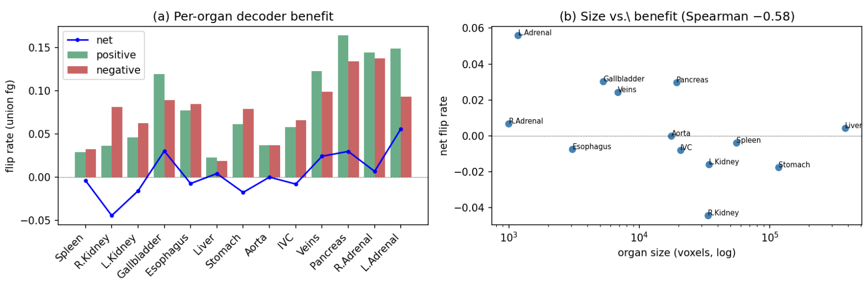


(f) MedNeXt_M Boundary Flips

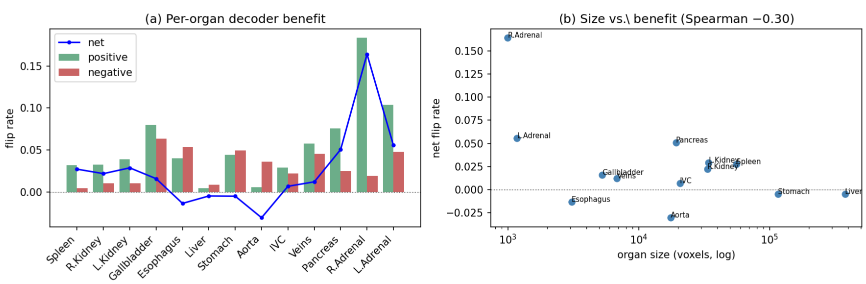
Figure 2: Cross-architecture voxel-level net benefit and spatial boundary concentration. Left Column: Global positive, negative, and net flip rates for (a) 3D UX-Net, (c) SwinUNETR, and (e) MedNeXt_M, demonstrating universal net-neutral behavior ($R_{net} \approx 0$). Right Column: Net flip rate by distance to organ boundaries for (b) 3D UX-Net, (d) SwinUNETR, and (f) MedNeXt_M, showing consistent concentration at 0–1 voxel boundary bands.



(a) 3D UX-Net Organ-Level Flips

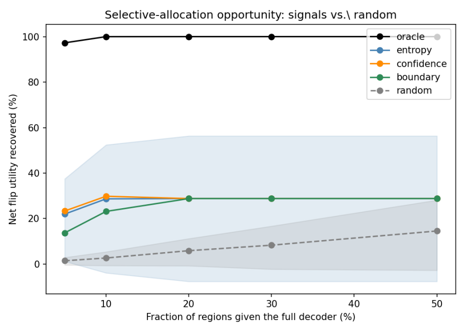


(b) SwinUNETR Organ-Level Flips

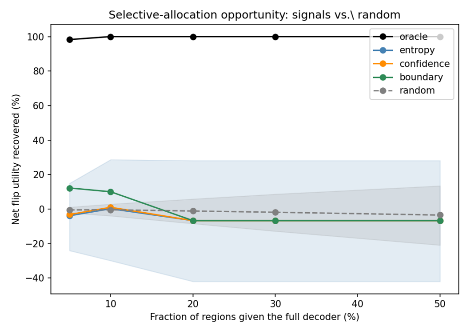


(c) MedNeXt_M Organ-Level Flips

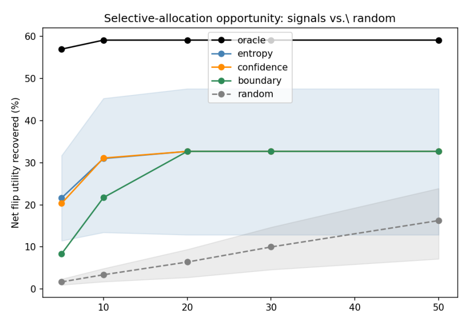
Figure 3: Organ-level heterogeneity comparison across all three backbones: (a) 3D UX-Net, (b) SwinUNETR, and (c) MedNeXt_M. Stacked vertically in full-width resolution, each panel clearly demonstrates that small and complex structures (e.g., adrenal glands and veins) exhibit high flip volatility, whereas large organs (e.g., liver) remain highly stable.



(a) 3D UX-Net Concentration Curve

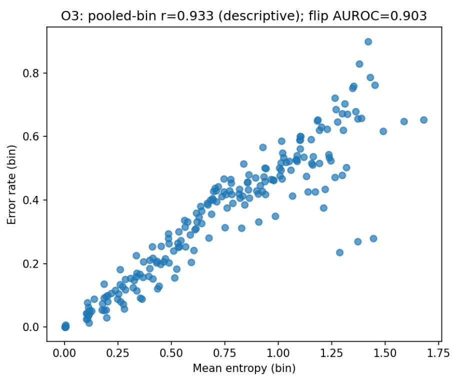


(b) SwinUNETR Concentration Curve

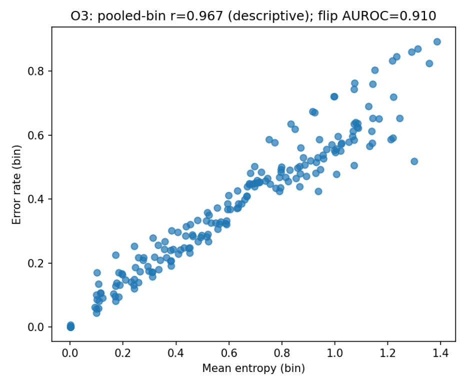


(c) MedNeXt_M Concentration Curve

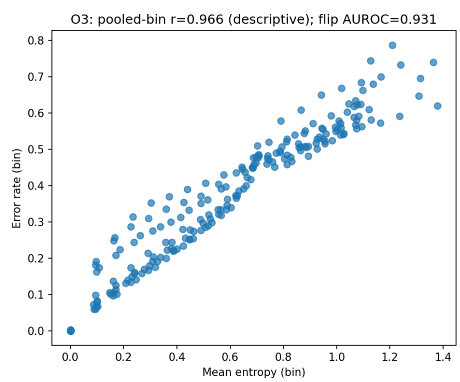
Figure 4: Oracle concentration curves across all three backbones: (a) 3D UX-Net, (b) SwinUNETR, and (c) MedNeXt_M. Across all networks, 80% of positive net benefit (K_{80}) is concentrated in just 5% of the volume.



(a) 3D UX-Net Predictability



(b) SwinUNETR Predictability



(c) MedNeXt_M Predictability

Figure 5: Inference-time flip predictability across all three backbones: (a) 3D UX-Net, (b) SwinUNETR, and (c) MedNeXt_M. Across all networks, entropy highly discriminates flip locations (instability, AUROC > 0.90) but fails to distinguish positive corrections from negative degradations (AUROC ≈ 0.55).